

Joel Hakavuori

Curriculum Vitae

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🌐 <https://hakav.github.io/>

Education

- 1/2025– **PhD Student**, *Sorbonne Université / IMJ-PRG*
Advised by Karim Adiprasito. On exchange for the 2025-26 academic year as a Fulbright grantee at the University of Michigan, advised by David Speyer.
- 8/2020– **BSc & MSc, Mathematics & Engineering physics**, *Aalto University*, GPA 4.94/5
9/2024 MSc thesis: "Nonpositive curvature, ℓ^2 -invariants, and right-angled Coxeter groups". Advised by Karim Adiprasito and Kaie Kubjas. Exchange semester at EPFL in 2023/24.

Experience

- 5/2024– **Research Assistant**, *Aalto University, Department of Mathematics*
7/2024 Research assistant in the group of Prof. Kaie Kubjas.
- 5/2023– **Visiting Research Fellow**, *Hebrew University of Jerusalem, Einstein Institute of Mathematics*
7/2023 REU participant, advised by Karim Adiprasito and Eran Nevo.
- 6/2022– **Research Assistant**, *Aalto University, Department of Mathematics*
8/2022 Research project advised by Alexander Engström and Milo Orlich.
- 9/2021– **Teaching Assistant**, *Aalto University, Department of Mathematics*
12/2024

Research

The g -theorem in smooth approximation, 2026, available at [arXiv:2609.24790](https://arxiv.org/abs/2609.24790)
Mixed Ehrhart Theory, 2026, available [here](#)
Subadditivity of shifts, Eilenberg-Zilber shuffle products and homology of lattices (with Karim Adiprasito, Anders Björner, Minas Margaritis and Volkmar Welker), 2024. Submitted, available at [arXiv:2404.16643](https://arxiv.org/abs/2404.16643).

Presentations

- October 2026 Queen Mary University of London, Combinatorial Algebraic Geometry Seminar
September 2026 IMJ-PRG, Combinatorics & Optimization Journée d'équipe
- August 2026 Aalto University, Algebra, Geometry and Combinatorics seminar
June 2026 AICove 2026, Poster session
- February 2026 University of Michigan Student combinatorics seminar
January 2026 University of Michigan Commutative algebra seminar
August 2026 Bernoulli Center, Lausanne, Summer School on Positivity in K-theory and Lattice Points

Honours and Awards

- 2024 **Fulbright Finland Foundation**, *Fulbright–Technology Industries of Finland Doctoral Grant*, \$40,000
Fulbright grant for 1-year doctoral research visit in the US.

- 2024 **Osk. Huttunen Foundation**, *Doctoral Grant*, 85,500€
A three-year grant for doctoral studies in mathematics at the Institut de Mathématiques de Jussieu-Paris Rive Gauche (IMJ-PRG).
- 2023 **Aalto University**, *Combined Student Fund Scholarship*, 1,000€
Award for academically successful master's students who have completed their BSc degree in Aalto University.
- 2022 **Aalto University**, *Professor E.J. Nyström's scholarship*, 1,000€
Award for success in studies in mathematics.